

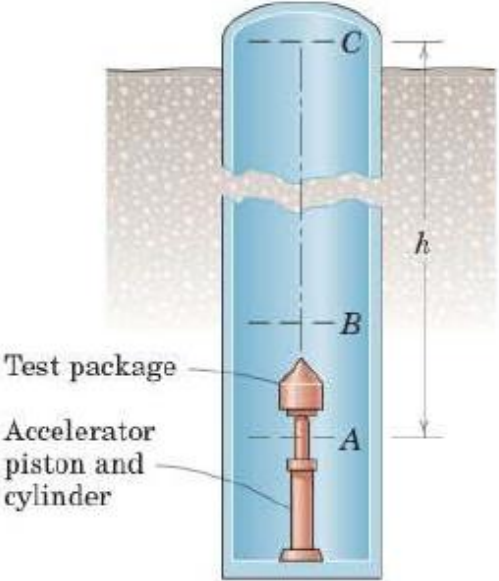
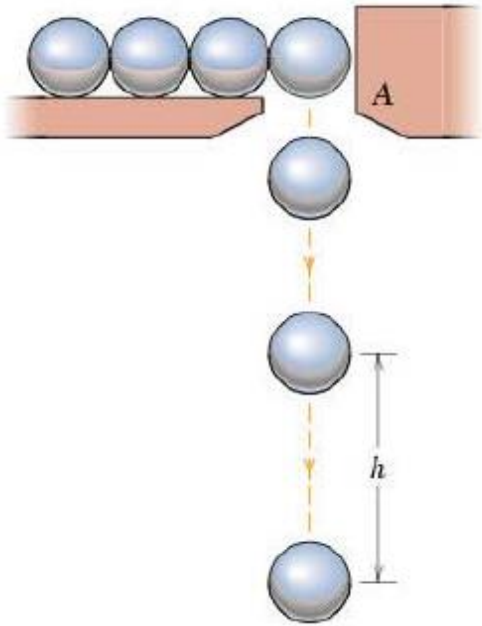
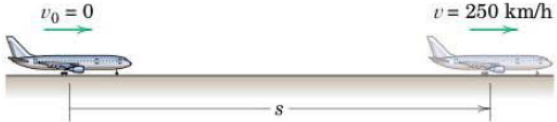
MEL1010 Engineering Mechanics

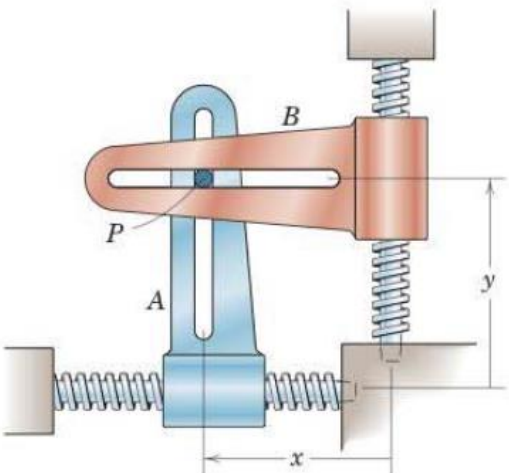
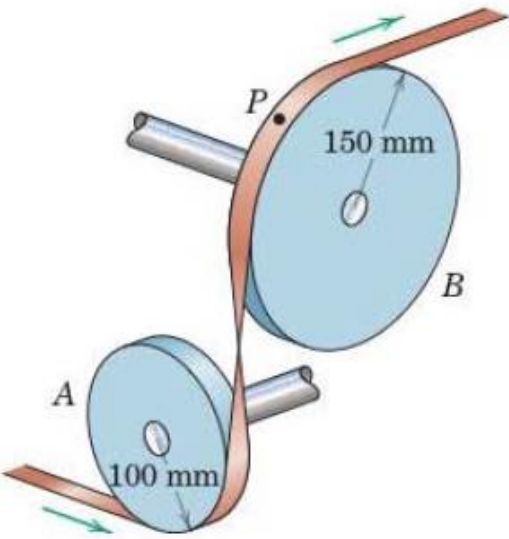
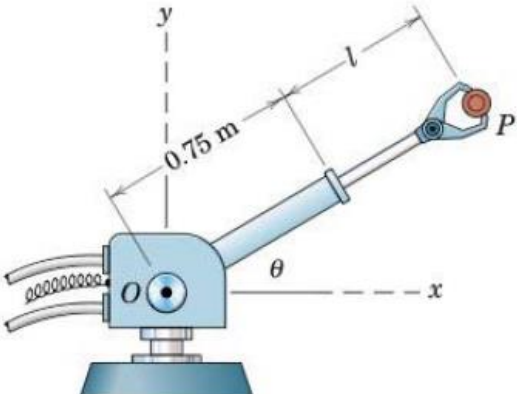
Tutorial 5 Kinematics of Particles

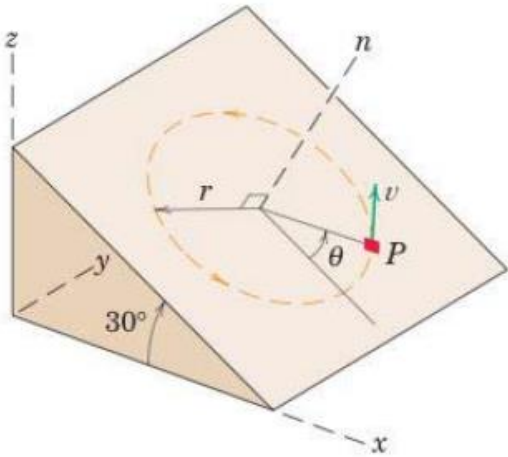
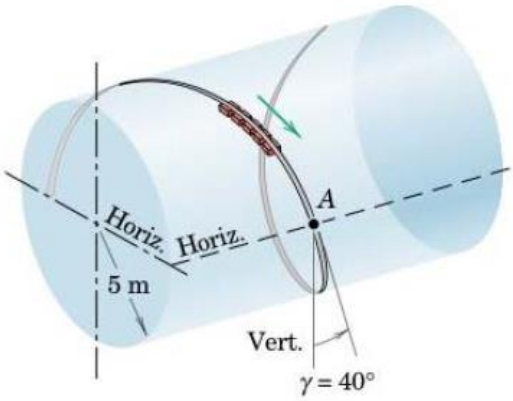
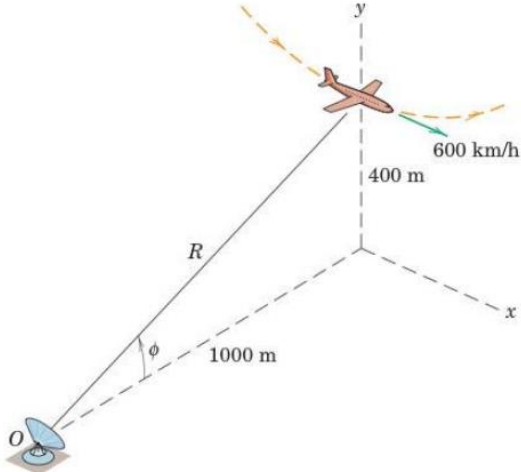
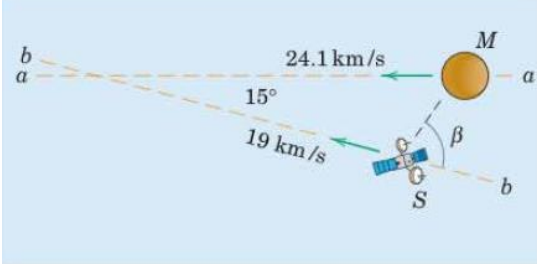
Tutorial Class: Thursday, 26th September 2024

Answer following questions. Mention valid assumptions wherever applicable

Questions about rectilinear motion, curvilinear motion using normal-tangential coordinates and polar coordinates, relative motion, space curvilinear motion, constrained motion

1	To test the effects of “weightlessness” for short periods of time, a test facility is designed which accelerates a test package vertically up from A to B by means of a gas-activated piston and allows it to ascend and descend from B to C to B under free-fall conditions. The test chamber consists of a deep well and is evacuated to eliminate any appreciable air resistance. If a constant acceleration of $40g$ from A to B is provided by the piston and if the total test time for the ‘weightless’ condition from B to C to B is 10 s, calculate the required working height h of the chamber. Upon returning to B, the test package is recovered in a basket filled with polystyrene pellets inserted in the line of fall. <i>Ans. $h = 125.7\text{ m}$</i>	 “Zero-g” test facility
2	Small steel balls fall from rest through the opening at A at the steady rate of two per second. Find the vertical separation h of two consecutive balls when the lower one has dropped 3 meters. Neglect air resistance. <i>Ans. $h = 2.61\text{ m}$</i>	
3	On its takeoff roll, the airplane starts from rest and accelerates according to $a = a_0 - kv^2$, where a_0 is the constant acceleration resulting from the engine thrust and $-kv^2$ is the acceleration due to aerodynamic drag. If $a_0 = 2\text{ m/s}^2$, $k =$	

	0.00004 m^{-1}, and v is in meters per second, determine the design length of runway required for the airplane to reach the take-off speed of 250 km/h if the drag term is (a) excluded and (b) included. <i>Ans. (a) $s = 1206 \text{ m}$, (b) $s = 1268 \text{ m}$,</i>	
4	The x- and y- motion of guides A and B with right-angle slots control the curvilinear motion of the connecting pin P, which slides in both slots. For a short interval, the motions are governed by $x = 20 + \frac{1}{4}t^2$ and $y = 15 - \frac{1}{6}t^3$, where x and y are in millimeters and t is in seconds. Calculate the magnitude of the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the pin for $t = 2 \text{ s}$. Sketch the direction of the path and indicate its curvature for this instant. <i>Ans. $v = 2.24 \text{ mm/s}$, $a = 2.06 \text{ mm/s}^2$</i> <i>Note: Use Rectangular Coordinates</i>	
5	A rocket is released at point A from jet aircraft flying horizontally at 1000 km/h at an altitude of 800 m. If the rocket thrust remains horizontal and gives the rocket a horizontal acceleration of $0.5g$, determine the angle θ from the horizontal to the line of sight to the target. <i>Ans. $\theta = 30^\circ$</i> <i>Note: Use Rectangular Coordinates</i>	
6	The direction of motion of a flat tape in a numerical-control device is changed by the two pulleys A and B shown. If the speed of the tape increases uniformly from 2 m/s to 18 m/s while 8 meters of tape pass over the pulleys, calculate the magnitude of the acceleration of point P on the tape in contact with pulley B at the instant when the tape speed is 3 m/s. <i>Ans. $a = 63.2 \text{ m/s}^2$</i> <i>Note: Use normal-tangential coordinate</i>	
7	The robot arm is elevating and extending simultaneously. At a given instant, $\theta = 30^\circ$, $\dot{\theta} = 10 \text{ deg/s} = \text{constant}$, $l = 0.5 \text{ m}$, $\dot{l} = 0.2 \text{ m/s}$, and $\ddot{l} = -0.3 \text{ m/s}^2$. Compute the magnitudes of the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the gripped part P. In addition, express $\mathbf{v}$ and $\mathbf{a}$ in terms of unit vectors $\mathbf{i}$ and $\mathbf{j}$. <i>Ans. $v = 0.296 \text{ m/s}$, $a = 0.345 \text{ m/s}^2$</i> $\mathbf{v} = 0.064 \mathbf{i} + 0.289 \mathbf{j} \text{ m/s}$ $\mathbf{a} = -0.328 \mathbf{i} - 0.1086 \mathbf{j} \text{ m/s}^2$ <i>Note: Use Polar Coordinate</i>	

8	The small block P travels with constant speed v in the circular path of radius r on the inclined surface. If $\theta = 0$ at time $t = 0$, determine the x-, y-, and z-components of velocity and acceleration as function of time. <i>Ans.</i> $x = \frac{\sqrt{3}}{2} r \cos \frac{v}{r} t, y = r \sin \frac{v}{r} t, z = -\frac{1}{2} r \cos \frac{v}{r} t$ $\dot{x} = -\frac{\sqrt{3}}{2} v \sin \frac{v}{r} t, \dot{y} = v \cos \frac{v}{r} t, \dot{z} = \frac{1}{2} v \sin \frac{v}{r} t$ $\ddot{x} = -\frac{\sqrt{3}}{2} \frac{v^2}{r} \cos \frac{v}{r} t, \ddot{y} = -\frac{v^2}{r} \sin \frac{v}{r} t, \ddot{z} = \frac{1}{2} \frac{v^2}{r} \cos \frac{v}{r} t$ Note: Space Curvilinear Motion, Rectangular coordinates	
9	An amusement ride called the “corkscrew” takes the passengers through the upside-down curve of a horizontal cylindrical helix. The velocity of the cars as they pass position A is 15 m/s, and the component of their acceleration measured along the tangent to the path is $g \cos \gamma$ at this point. The effective radius of the cylindrical helix is 5 m, and the helix angle is $\gamma = 40^\circ$. Compute the magnitude of the acceleration of the passengers as they pass position A. <i>Ans.</i> $a = 27.5 \text{ m/s}^2$ Note: Space Curvilinear Motion, cylindrical coordinates	
10	At the bottom of a vertical loop in the x-y plane at an altitude of 400 m, the airplane has a speed of 600 km/h with no horizontal x-acceleration. The radius of curvature of the loop at the bottom is 1200 m. For the radar tracking at O, determine the recorded value of $\ddot{R}$ and $\ddot{\phi}$ for this instant. <i>Ans.</i> $\ddot{R} = 34.4 \text{ m/s}^2, \ddot{\phi} = 0.01038 \text{ rad/s}^2$ Note: Space Curvilinear Motion, Spherical Coordinates	
11	The spacecraft S approaches the planet Mars along a trajectory $b - b$ in the orbital plane of Mars with an absolute velocity of 19 km/s. Mars has a velocity of 24.1 km/s along its trajectory $a - a$. Determine the angle β between the line of sight $S - M$ and the trajectory $b - b$ when Mars appears from the spacecraft to be approaching it head on. <i>Ans.</i> $\beta = 55.6^\circ$ Note: Relative Motion	

	Optional	
12	Under the action of force P, the constant acceleration of block B is 3 m/s^2 to the right. At the instant when the velocity of B is 2 m/s to the right, determine the velocity of B relative to A, the acceleration of B relative to A, and the absolute velocity of point C of the cable. <i>Ans. $v_{B/A} = 0.5 \text{ m/s}$, $a_{B/A} = 0.75 \text{ m/s}^2$</i> <i>$v_C = 1 \text{ m/s}$, all to the right</i> Note: Constrained Motion of Connected particles	